

Matplotlib Demonstration

```
In [ ]: import matplotlib
```

```
In [ ]: print(matplotlib.__version__)
```

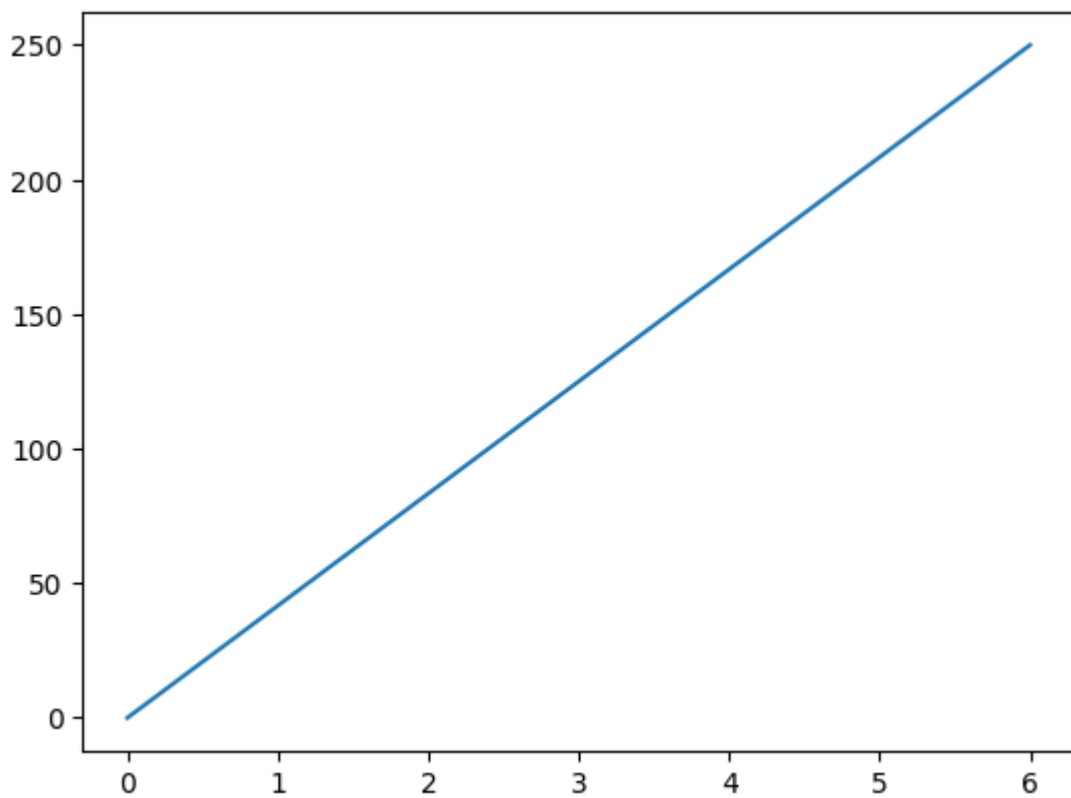
3.8.2

The usual way of importing

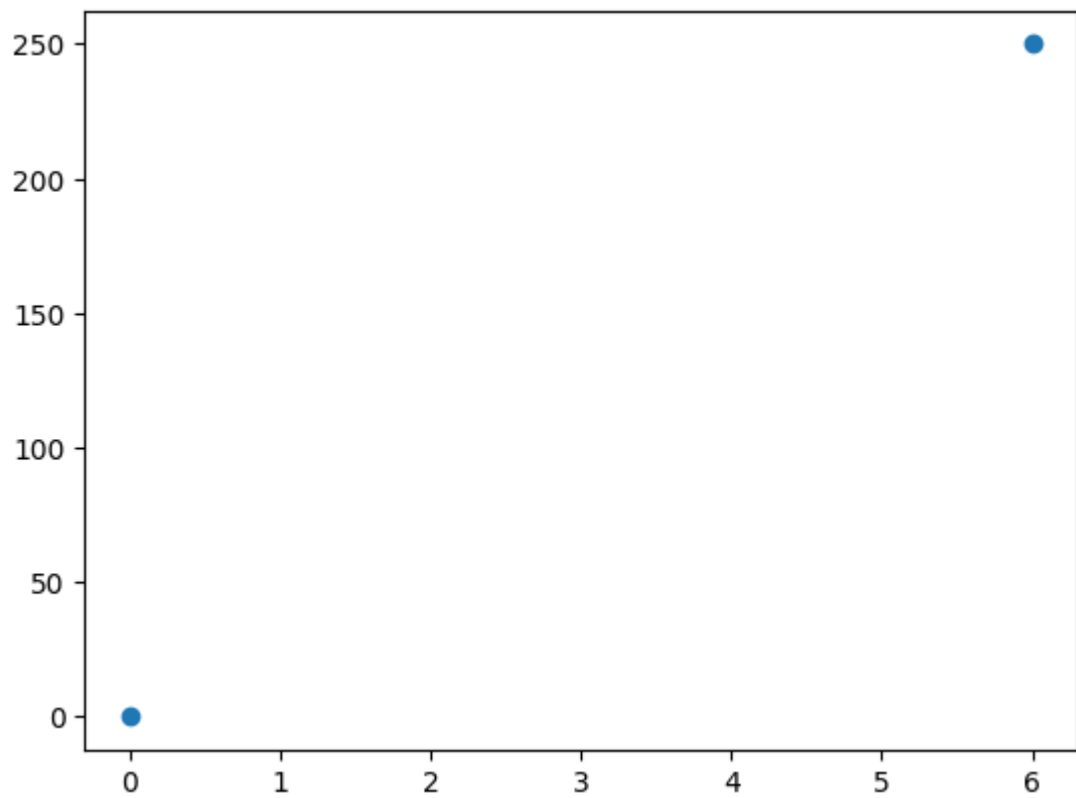
```
In [ ]: import matplotlib.pyplot as plt
```

```
In [ ]: import numpy as np
```

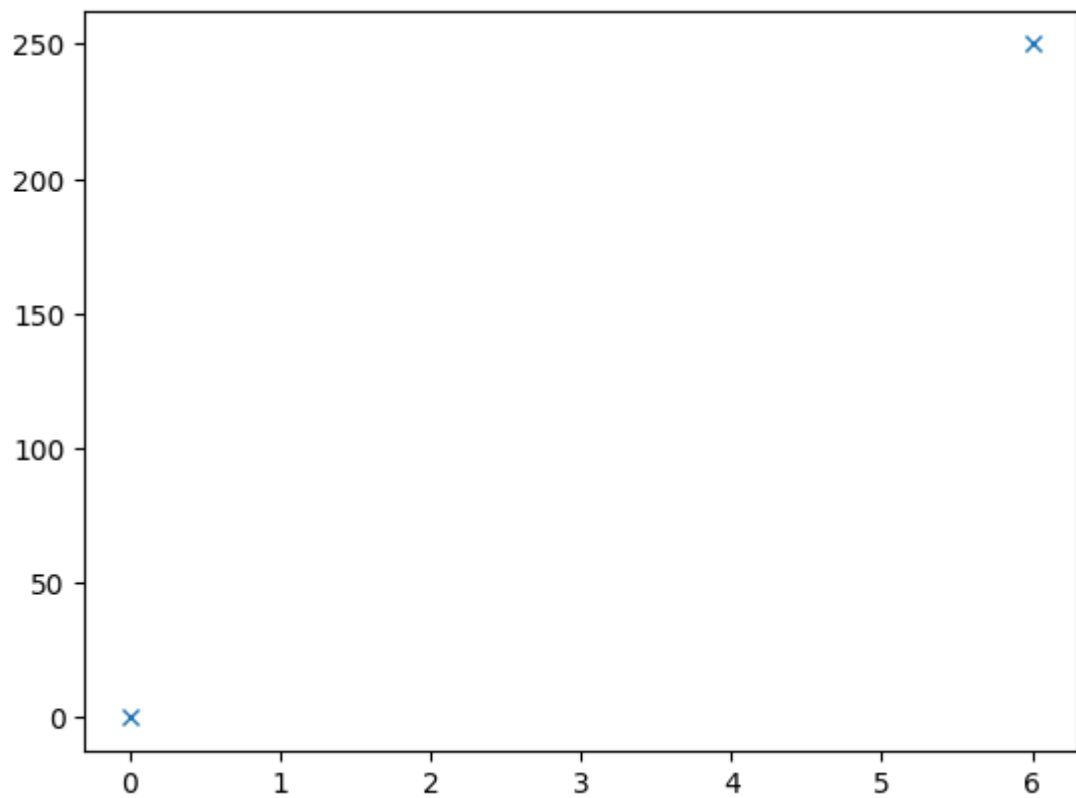
```
In [ ]: x_points = np.array([0,6])  
y_points = np.array([0,250])  
  
plt.plot(x_points, y_points)  
plt.show()
```



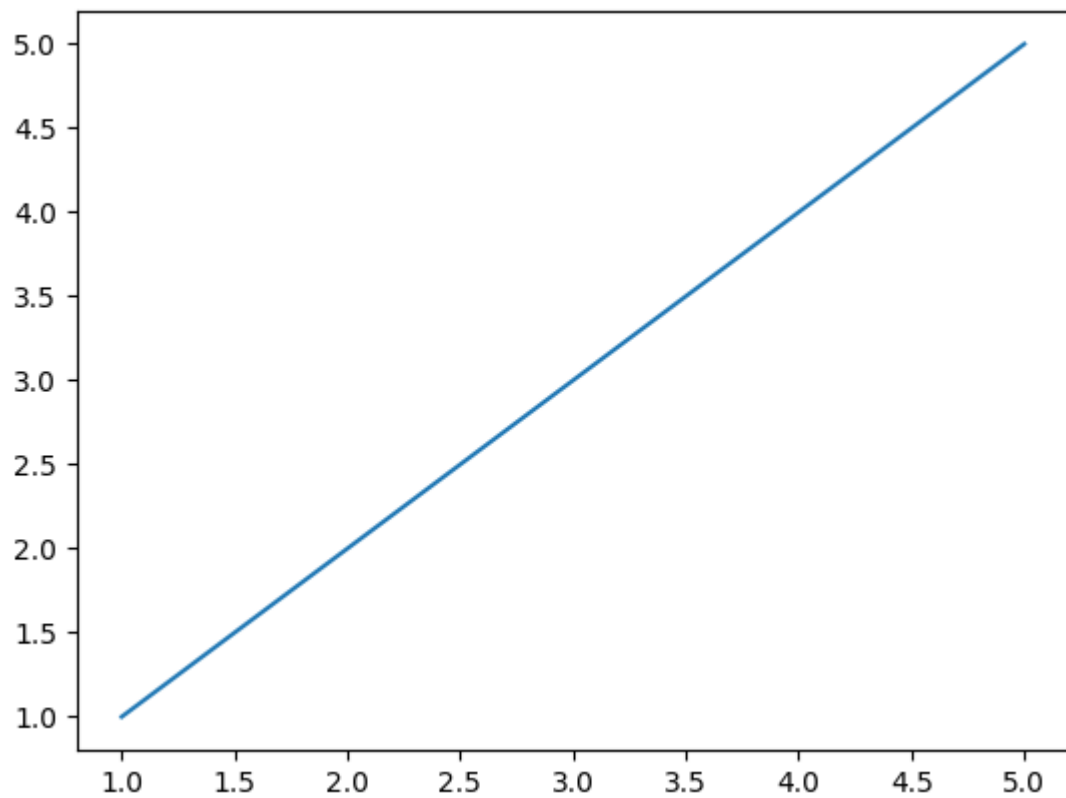
```
In [ ]: plt.plot(x_points, y_points, 'o')  
plt.show()
```



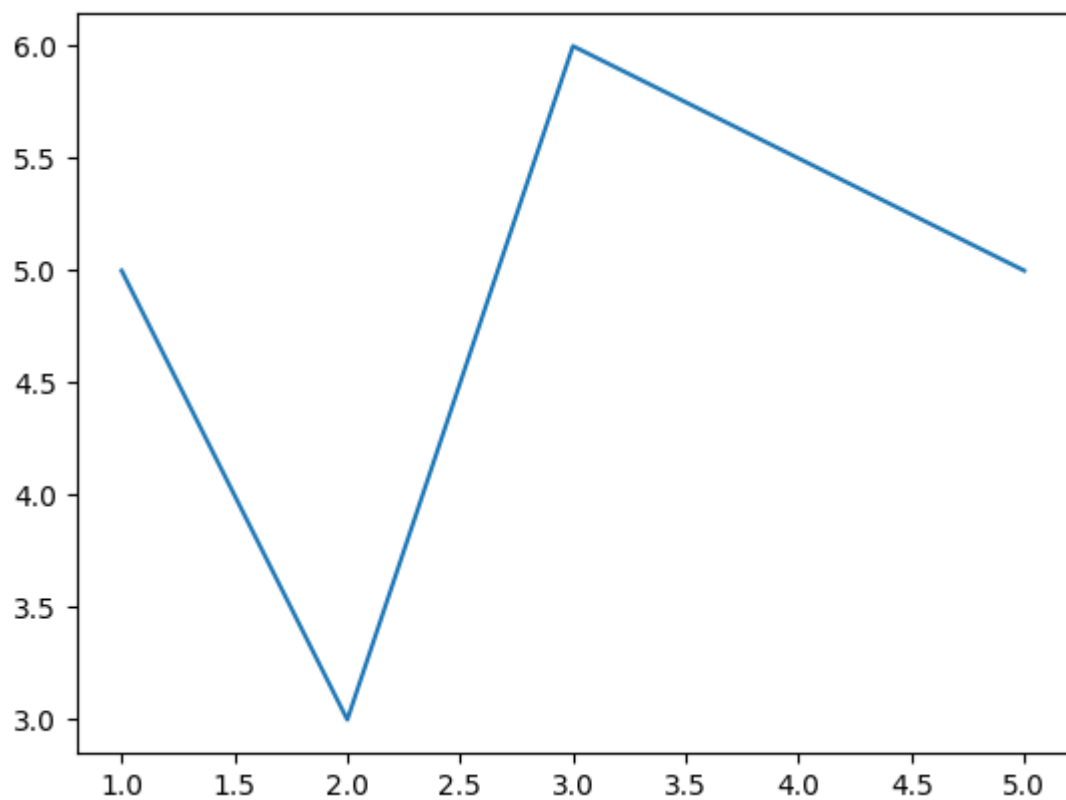
```
In [ ]: plt.plot(x_points, y_points, 'x')  
plt.show()
```



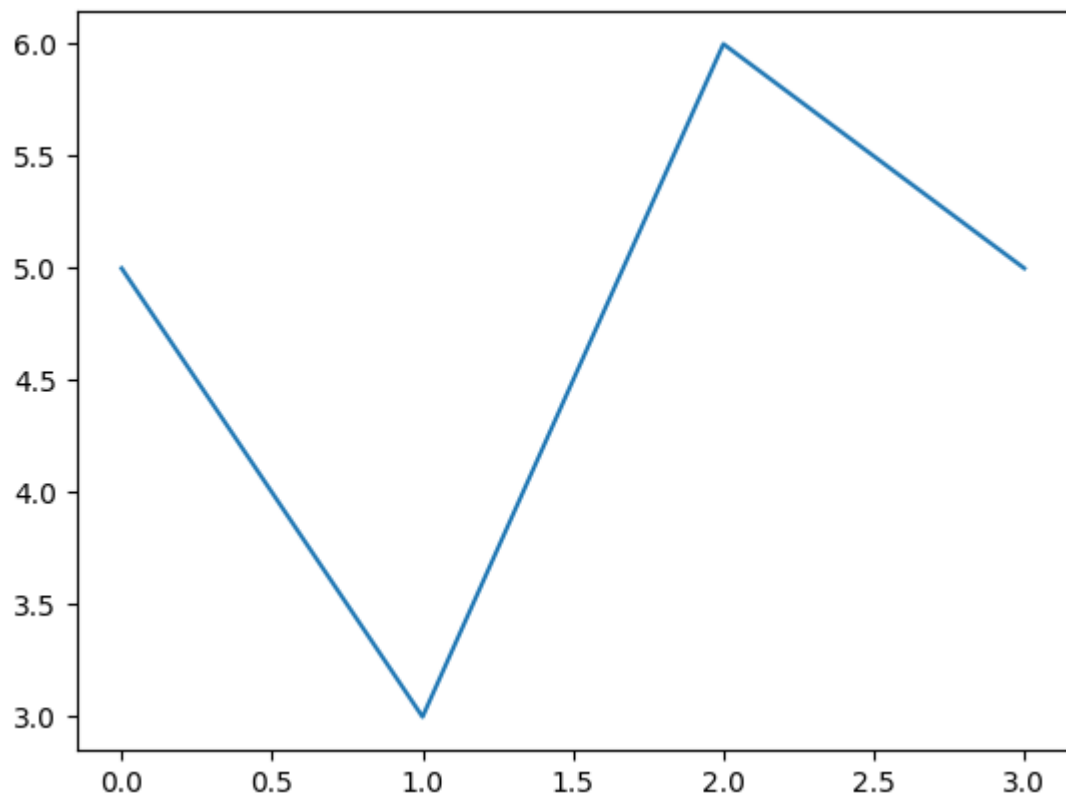
```
In [ ]: x_points = np.array([1,2,3,5])  
y_points = np.array([1,2,3,5])  
plt.plot(x_points, y_points)  
plt.show()
```



```
In [ ]: x_points = np.array([1,2,3,5])  
        y_points = np.array([5,3,6,5])  
        plt.plot(x_points, y_points)  
        plt.show()
```

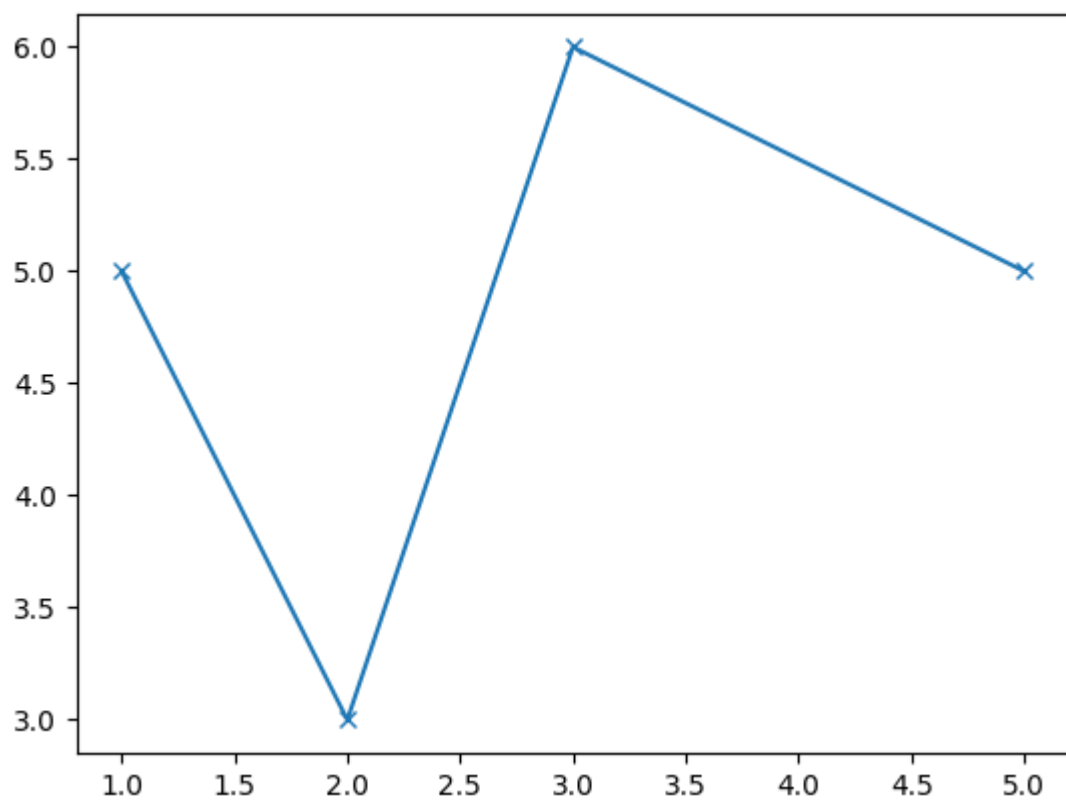


```
In [ ]: plt.plot(y_points)  
        plt.show()
```

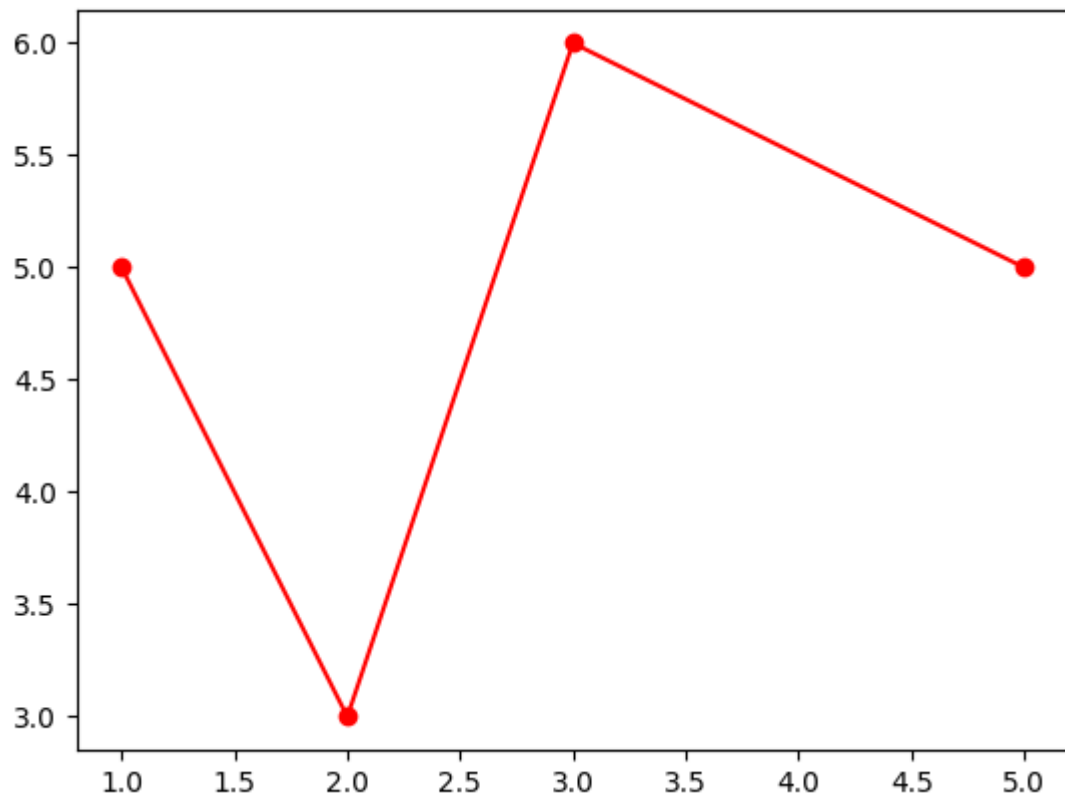


Properties

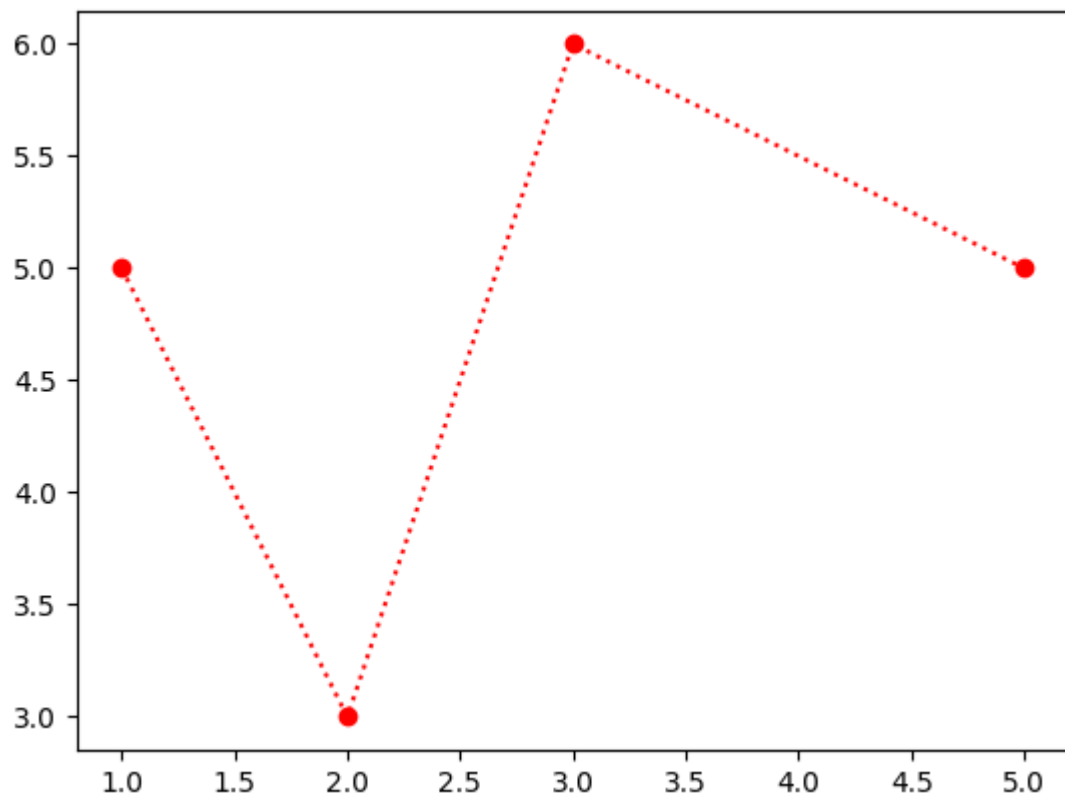
```
In [ ]: # Markers - 'o', 'x', '*'.....  
x_points = np.array([1,2,3,5])  
y_points = np.array([5,3,6,5])  
plt.plot(x_points, y_points, marker='x')  
plt.show()
```



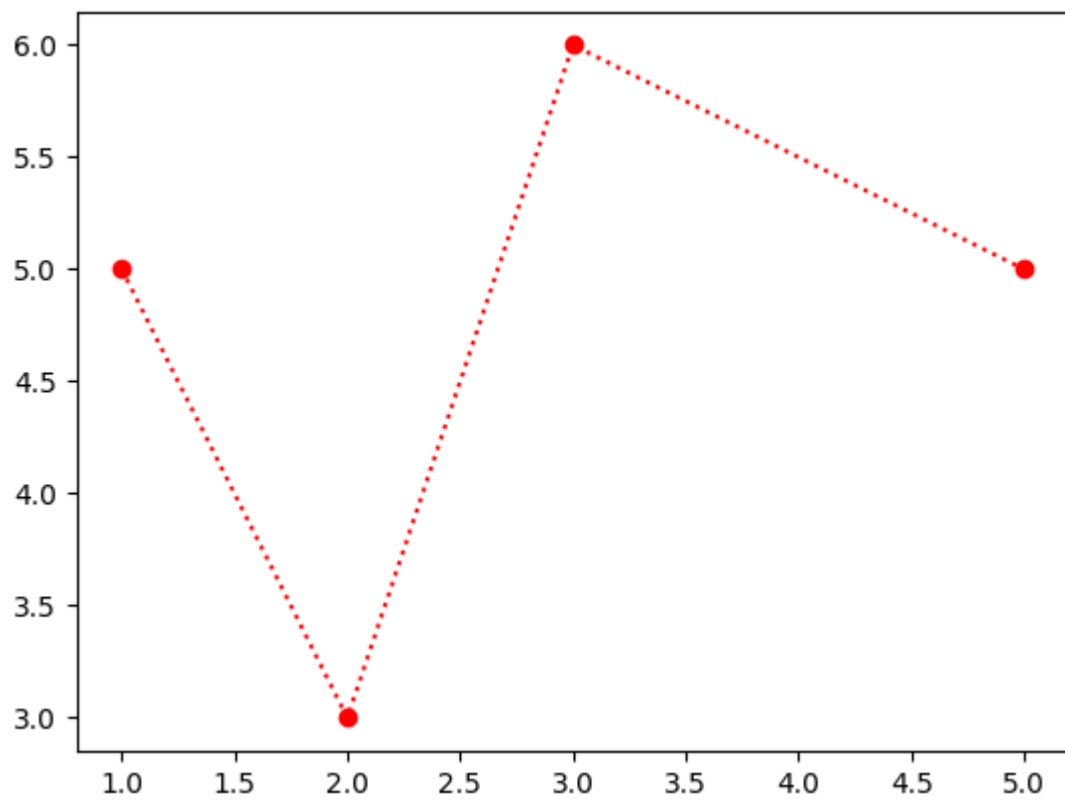
```
In [ ]: # Line Color
x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='r')
plt.show()
```



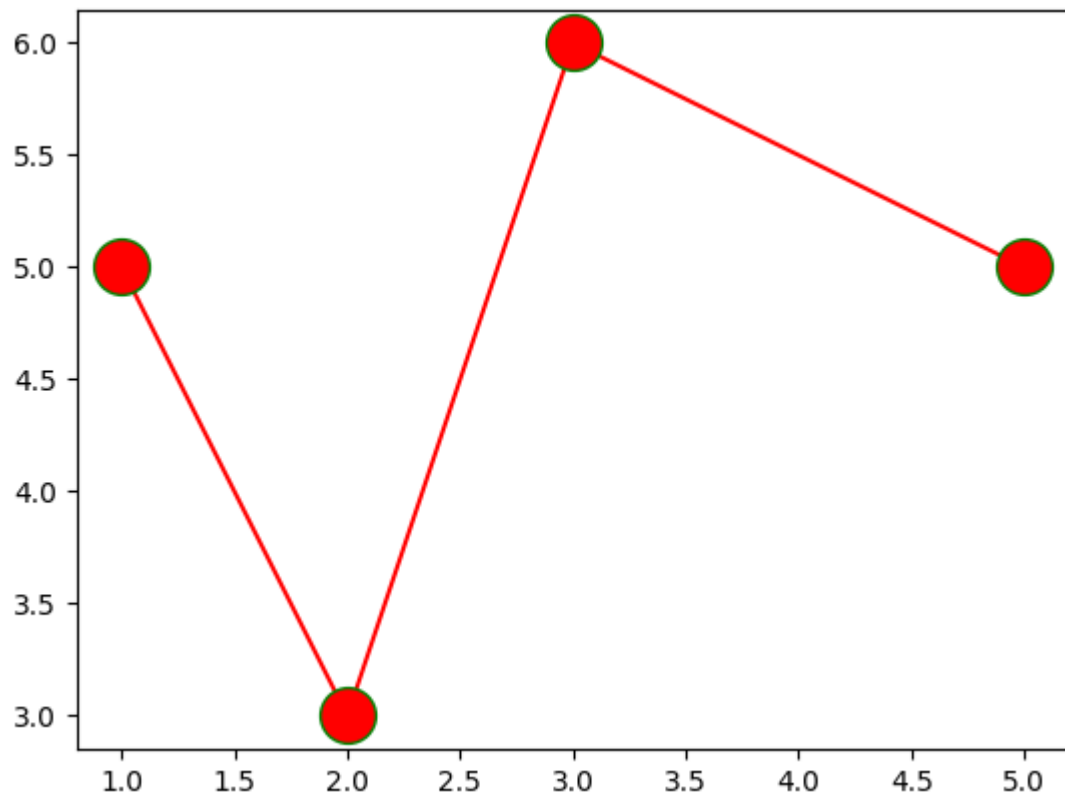
```
In [ ]: # Line style
x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='r', ls=':')
plt.show()
```



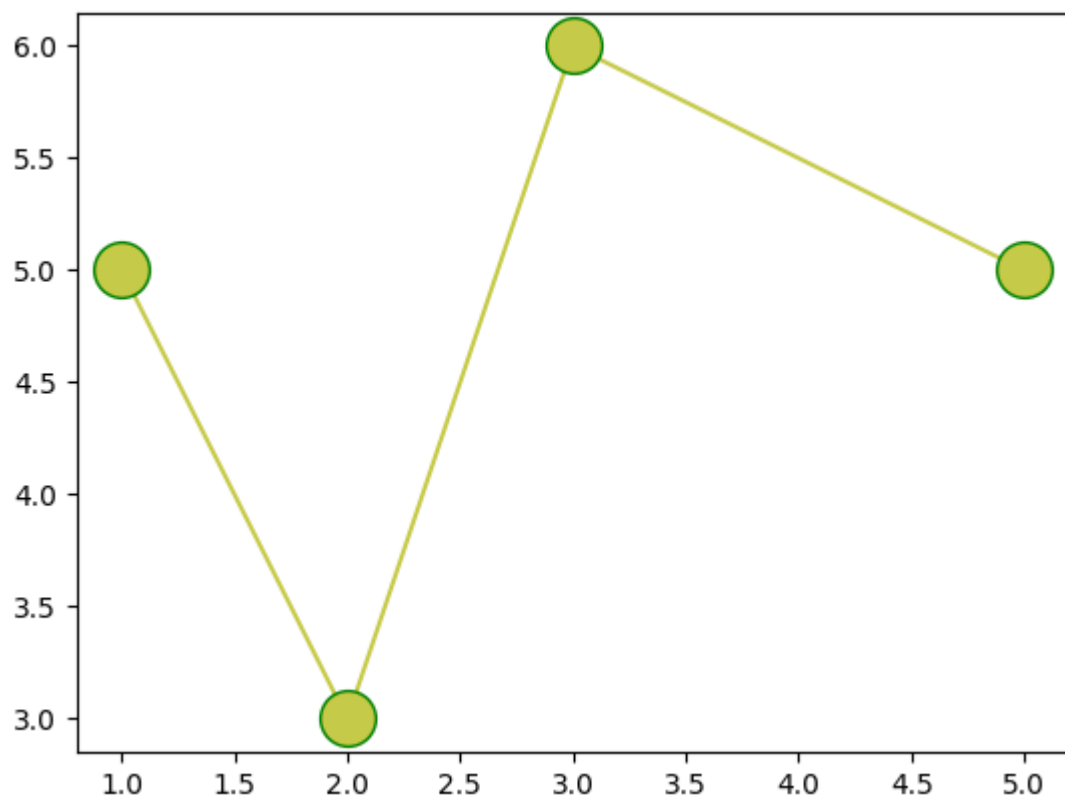
```
In [ ]: # Combined properties of marker|line|color
x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, 'o:r')
plt.show()
```



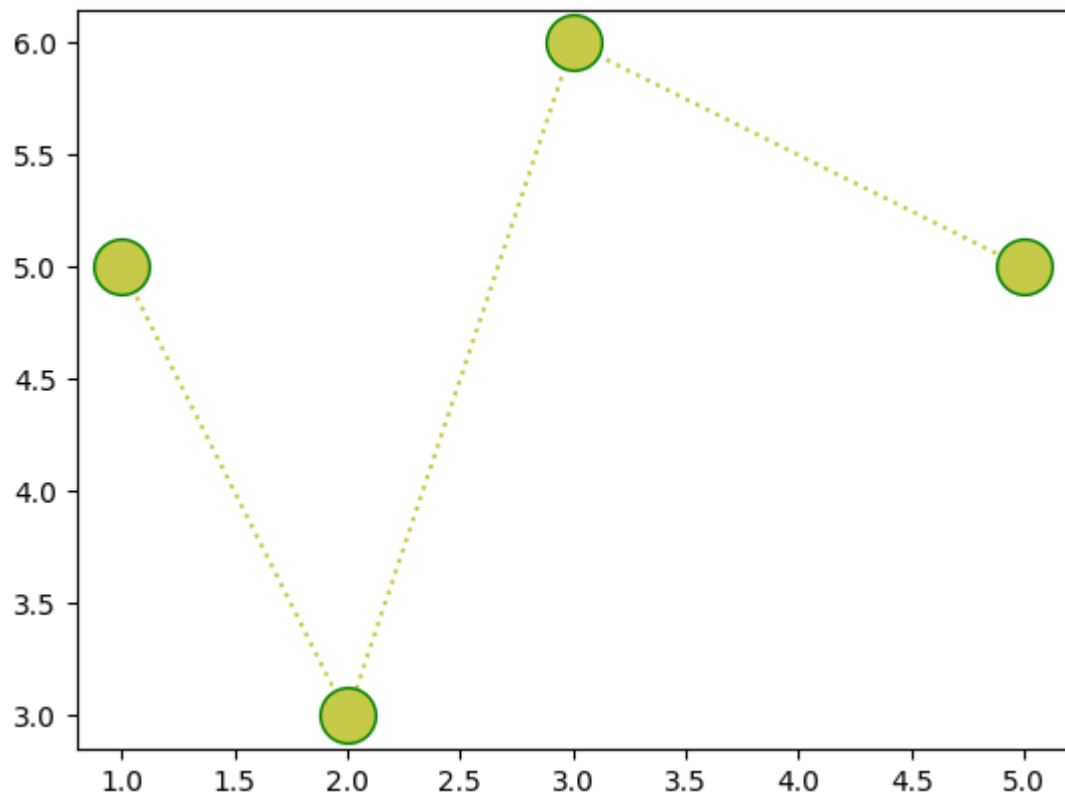
```
In [ ]: x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='r', mec='g', ms=20)
plt.show()
```



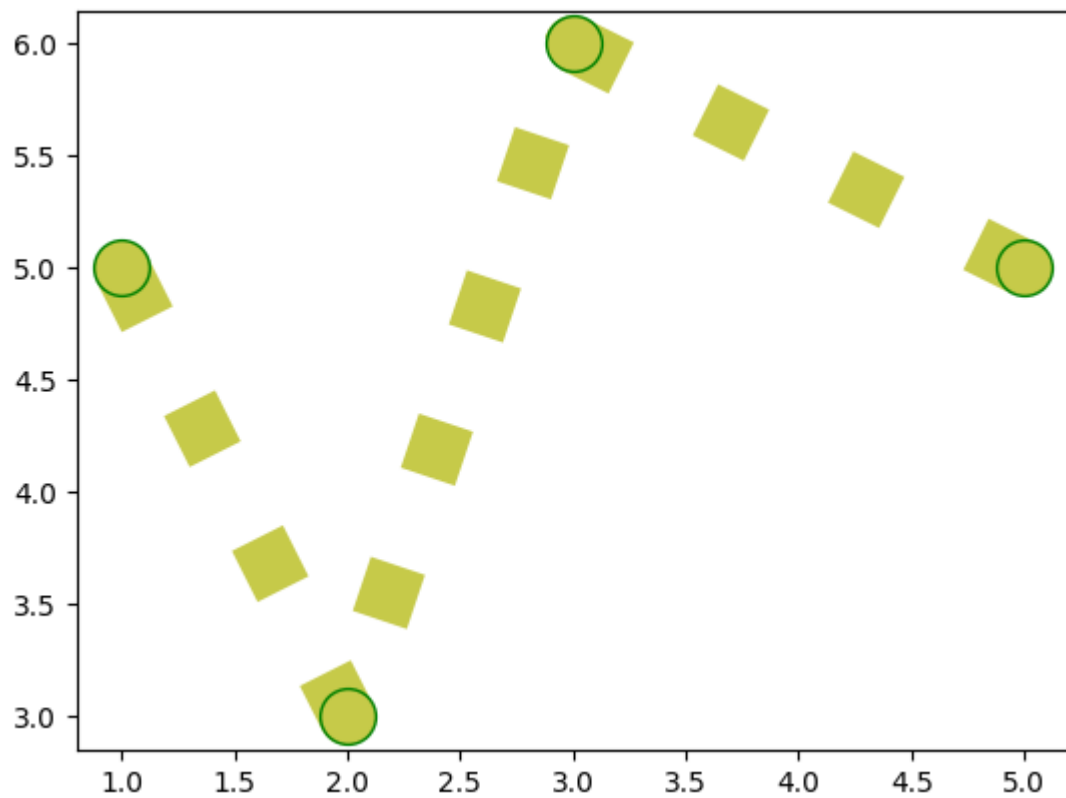
```
In [ ]: x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='#c6ca49', mec='g', ms=20)
plt.show()
```



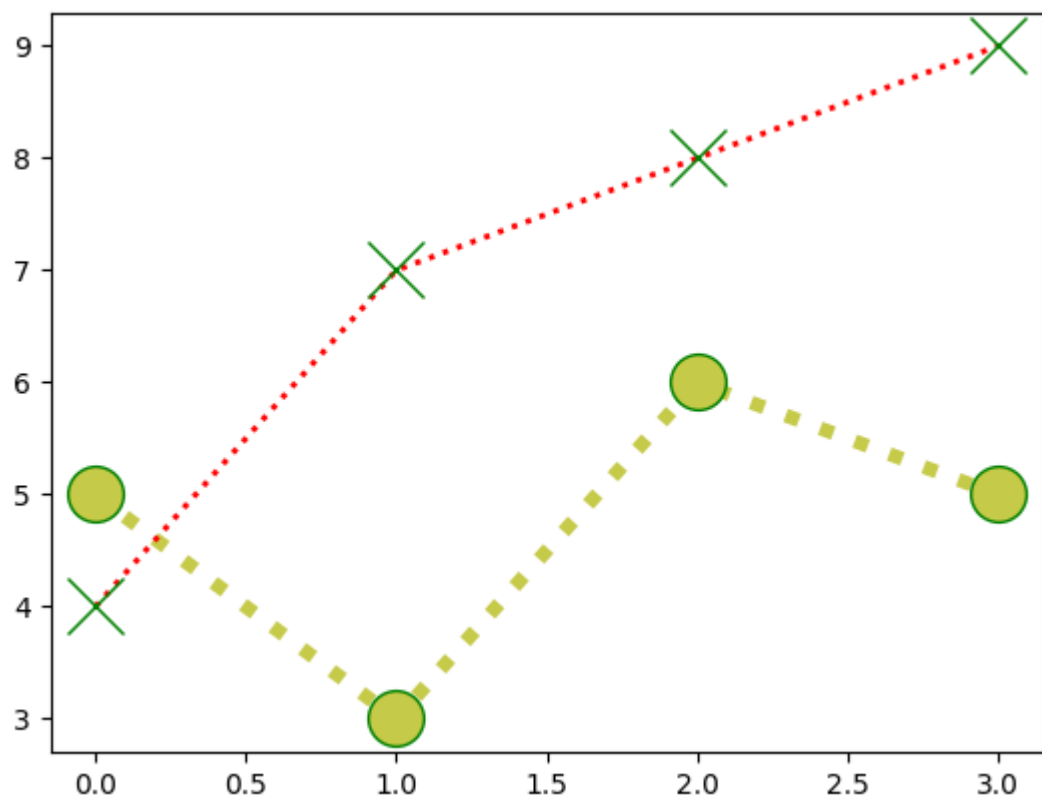
```
In [ ]: x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='#c6ca49', mec='g', ms=20,
plt.show())
```



```
In [ ]: # Line width
x_points = np.array([1,2,3,5])
y_points = np.array([5,3,6,5])
plt.plot(x_points, y_points, marker='o', color='#c6ca49', mec='g', ms=20,
plt.show())
```



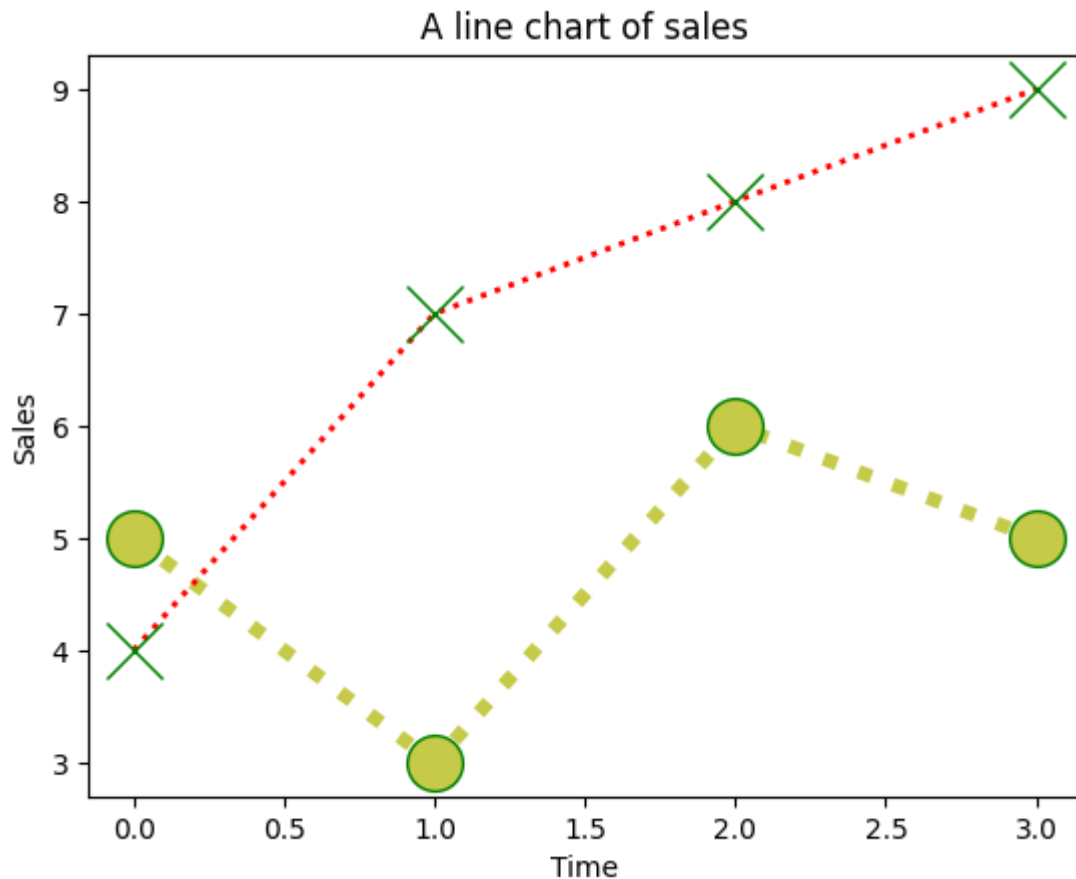
```
In [ ]: # Line width
y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])
plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',
plt.show()
```



Properties outside figure

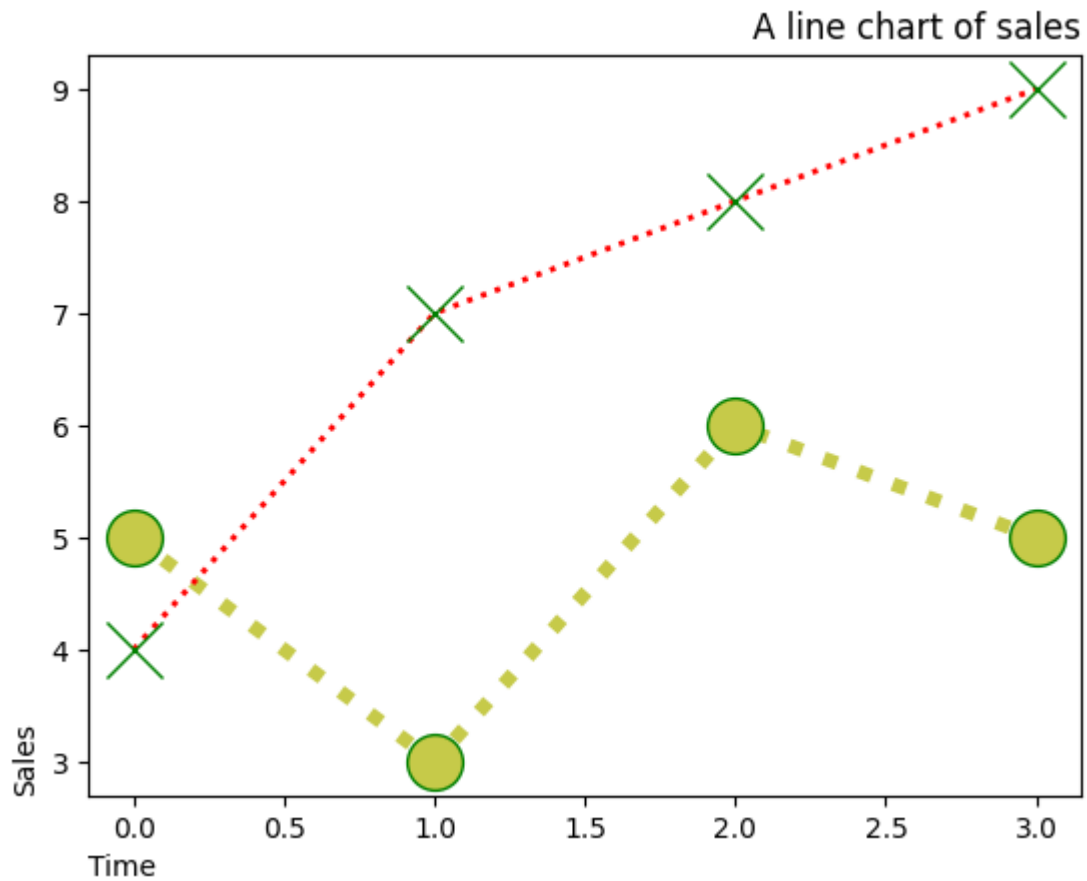
```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])
plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time")
plt.ylabel("Sales")
plt.title("A line chart of sales")
plt.show()
```



```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])
plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time", loc="left")
plt.ylabel("Sales", loc="bottom")
plt.title("A line chart of sales", loc="right")
plt.show()
```



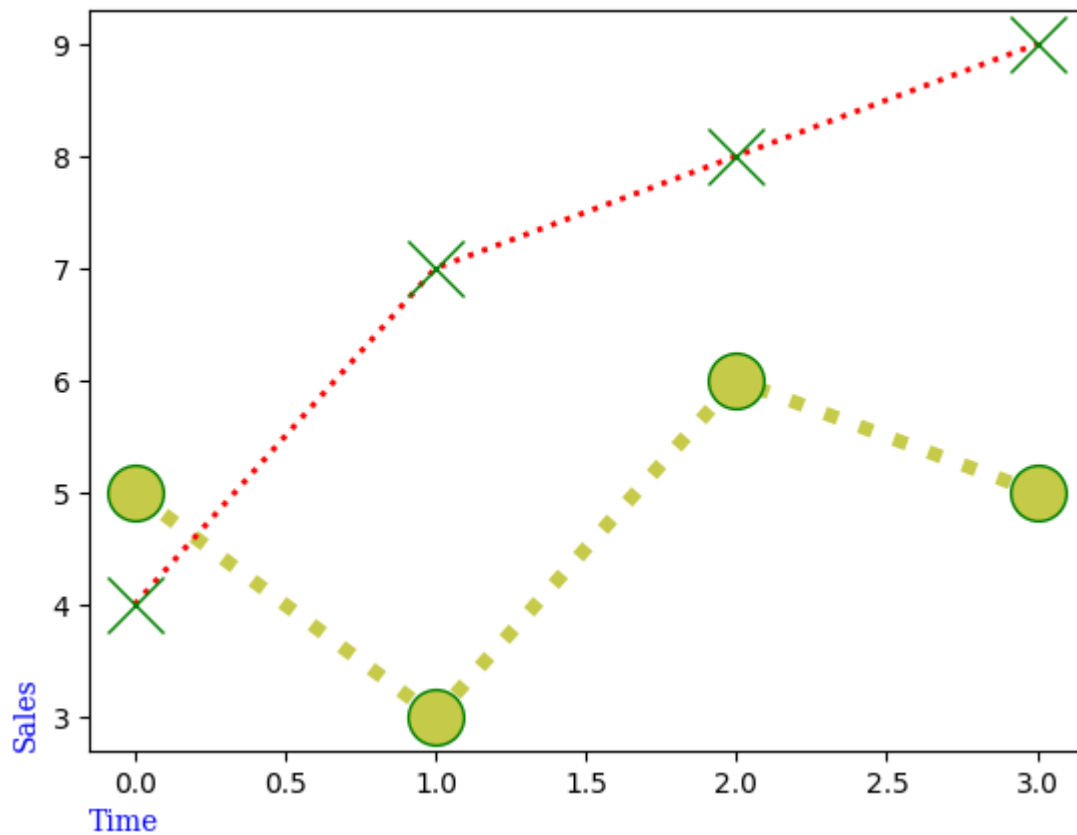
```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])

fontdict1 = {'family':"serif","color":"blue","size":10}
fontdict2 = {'family':"serif","color":"darkred","size":25}

plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.title("A line chart of sales",fontdict=fontdict2, loc="right")
plt.show()
```

A line chart of sales



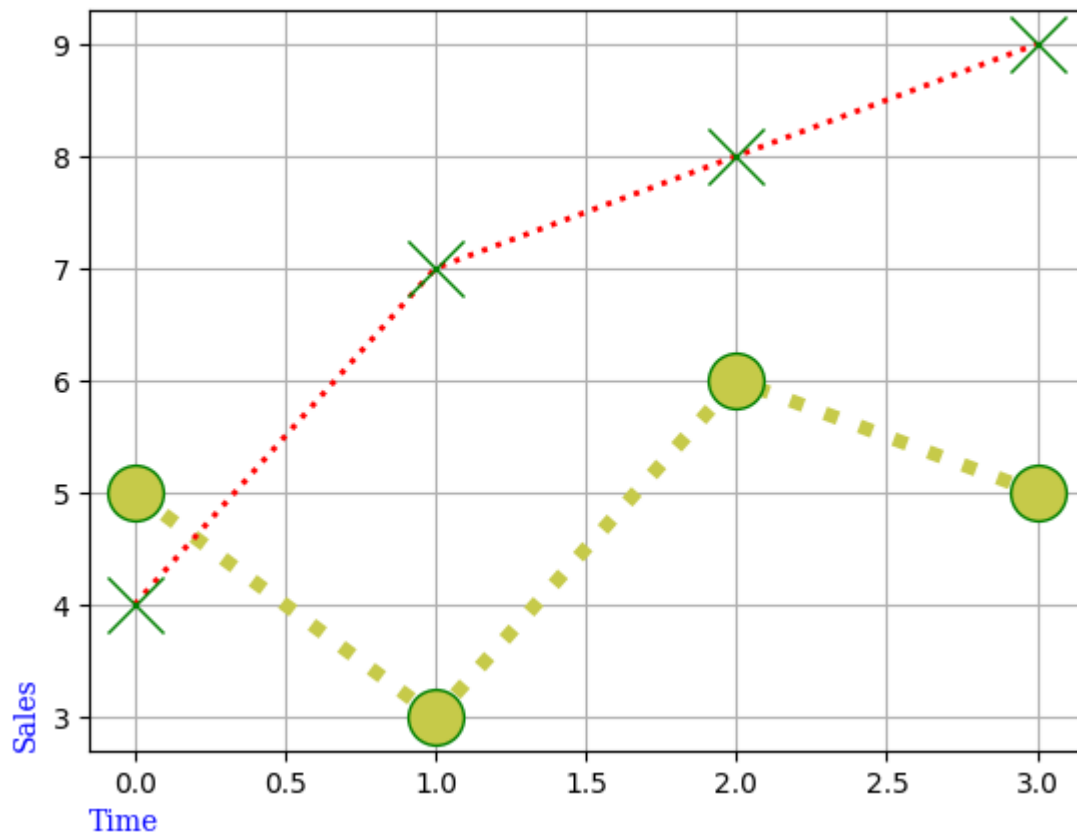
```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])

fontdict1 = {'family':'serif','color':'blue','size':10}
fontdict2 = {'family':'serif','color':'darkred','size':25}

plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.title("A line chart of sales",fontdict=fontdict2, loc="right")
plt.grid()
plt.show()
```

A line chart of sales



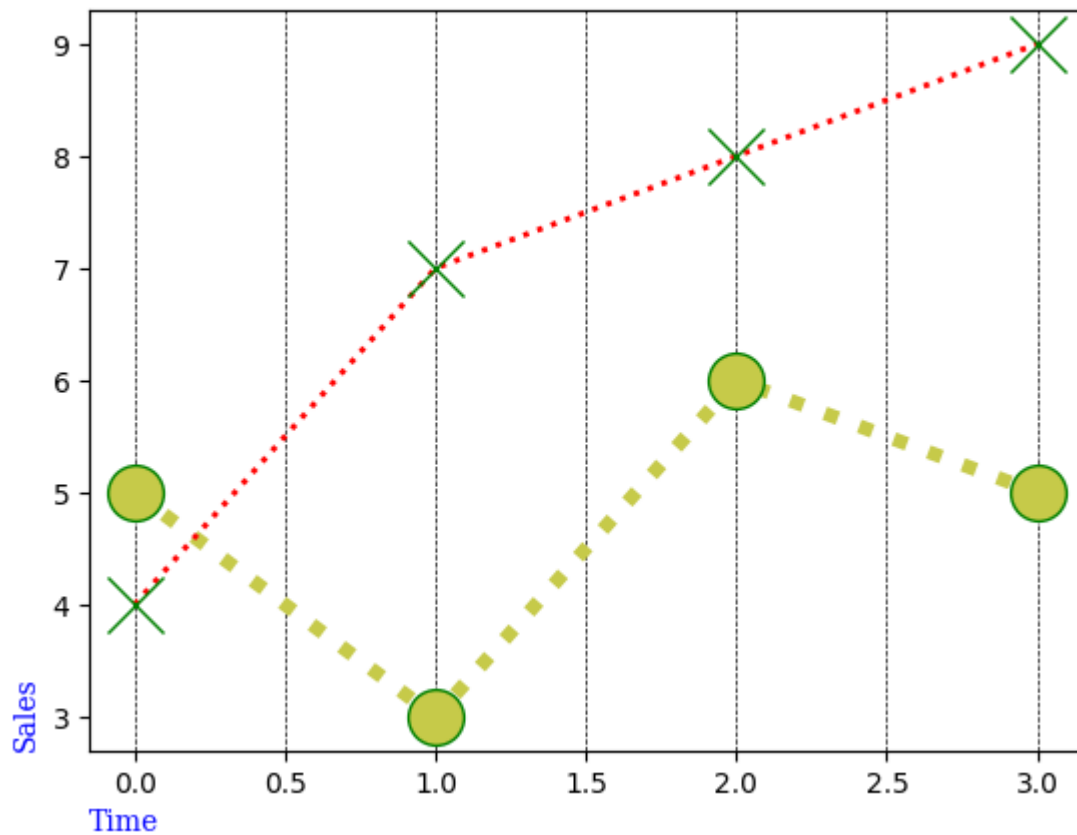
```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])

fontdict1 = {'family':"serif","color":"blue","size":10}
fontdict2 = {'family':"serif","color":"darkred","size":25}

plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.title("A line chart of sales",fontdict=fontdict2, loc="right")
plt.grid(axis='x', color='black', ls='--', lw='0.5')
plt.show()
```

A line chart of sales



Multple plots

```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])

fontdict1 = {'family':"serif","color":"blue","size":10}
fontdict2 = {'family':"serif","color":"darkred","size":25}

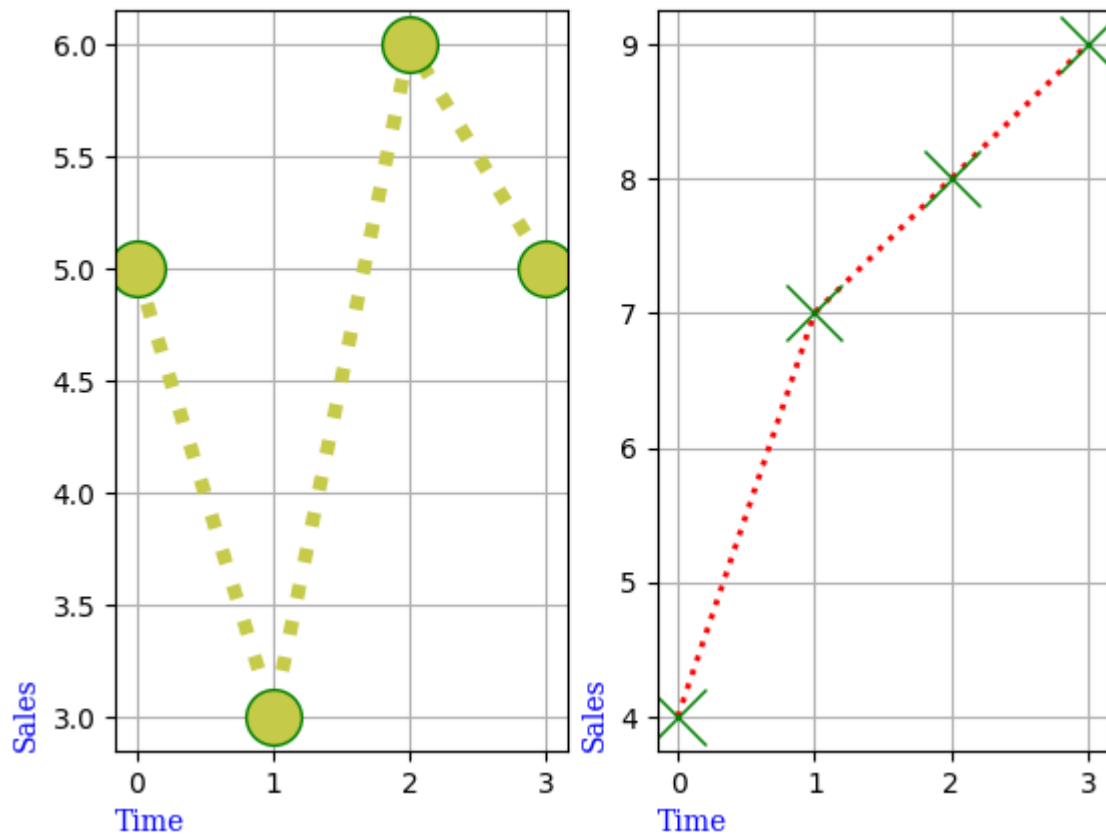
plt.subplot(1,2,1)
plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.grid()

plt.subplot(1,2,2)
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.title("A line chart of sales",fontdict=fontdict2, loc="right")
plt.grid()

plt.show()
```

A line chart of sales



```
In [ ]: y_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])

fontdict1 = {'family':"serif","color":"blue","size":10}
fontdict2 = {'family':"serif","color":"darkred","size":25}

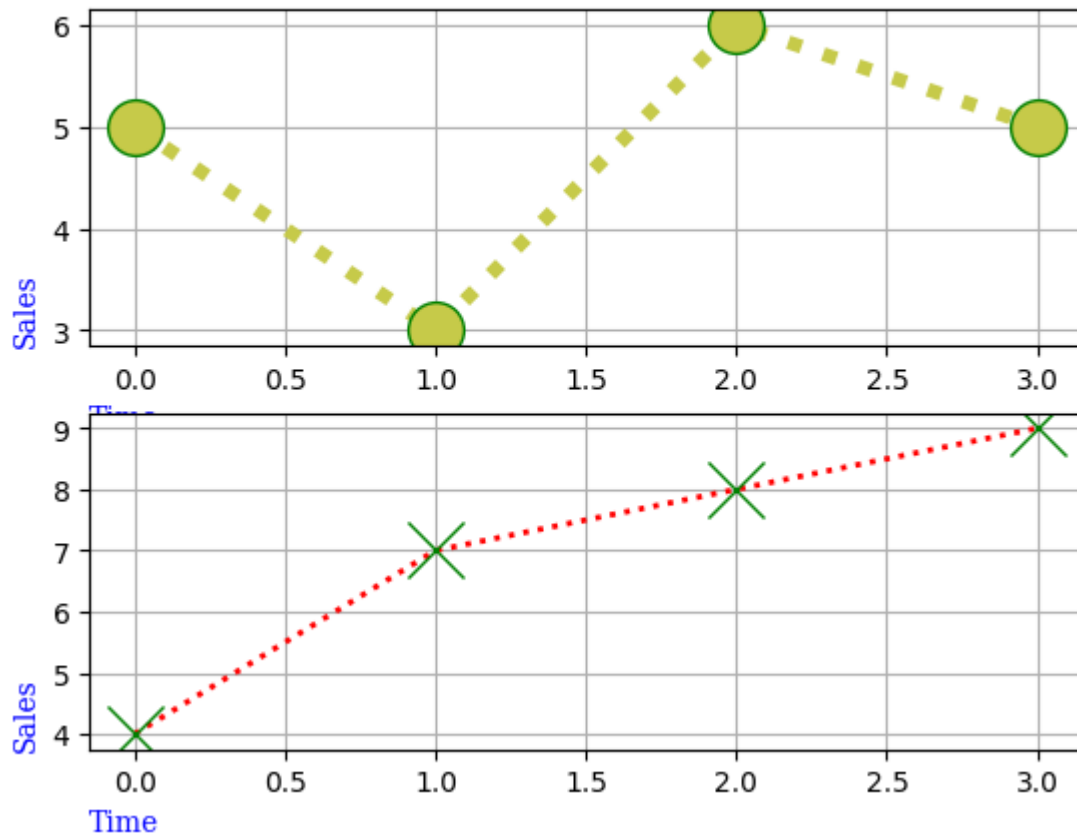
plt.subplot(2,1,1)
plt.plot(y_points, marker='o', color='#c6ca49', mec='g', ms=20, ls='dotted')
plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.grid()

plt.subplot(2,1,2)
plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

plt.xlabel("Time",fontdict=fontdict1, loc="left")
plt.ylabel("Sales",fontdict=fontdict1, loc="bottom")
plt.grid()

plt.suptitle("A line chart of sales",fontdict=fontdict2)
plt.show()
```

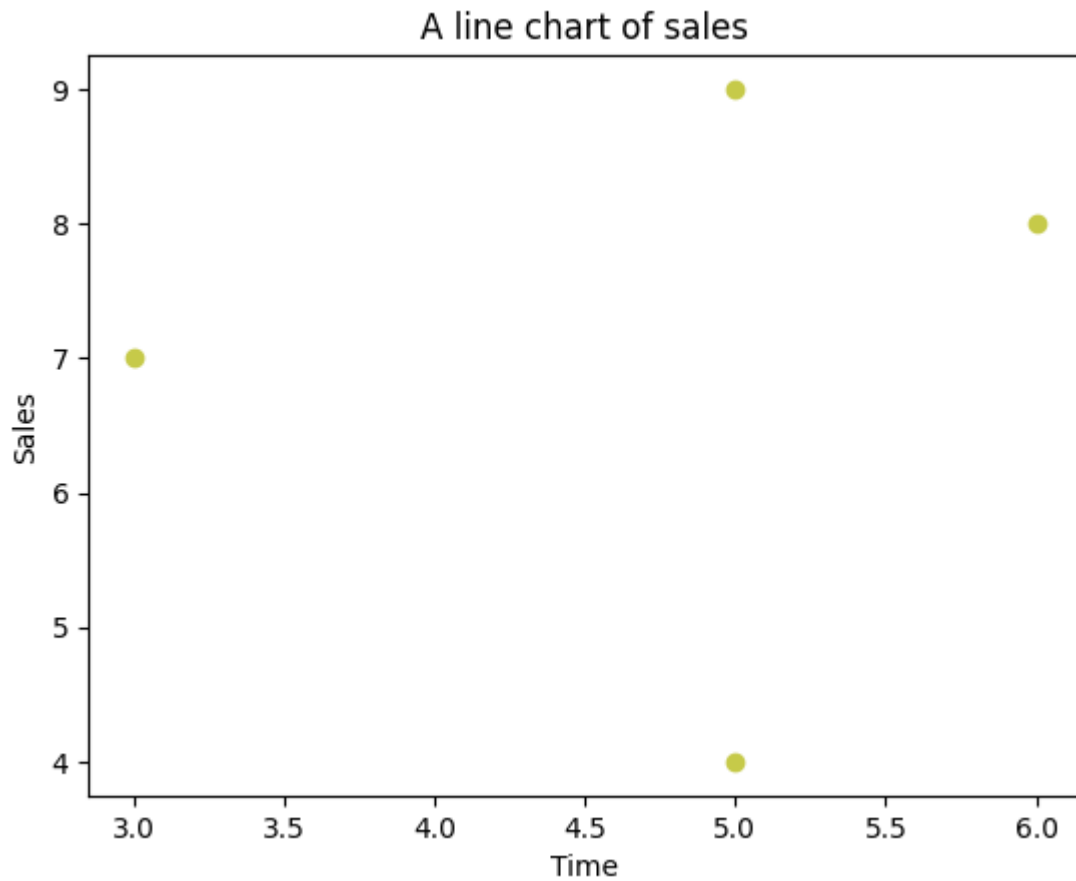
A line chart of sales



Scatter Plot

```
In [ ]: x_points = np.array([5,3,6,5])
y_points12 = np.array([4,7,8,9])
plt.scatter(x_points,y_points12, marker='o', color='#c6ca49')
#plt.plot(y_points12, marker='x', color='r', mec='g', ms=20, ls='dotted',

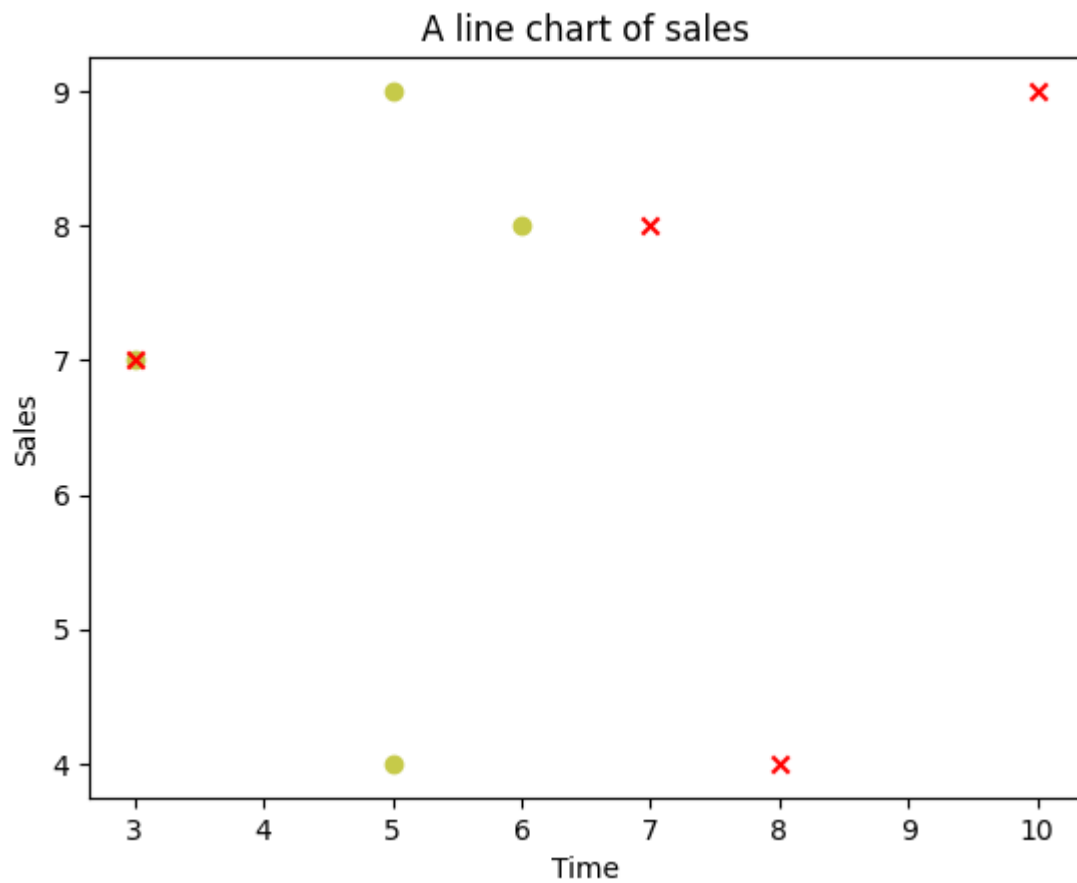
plt.xlabel("Time")
plt.ylabel("Sales")
plt.title("A line chart of sales")
plt.show()
```



```
In [ ]: x_points = np.array([5,3,6,5])
        y_points = np.array([4,7,8,9])

        x_points12 = np.array([8,3,7,10])
        y_points12 = np.array([4,7,8,9])
        plt.scatter(x_points,y_points, marker='o', color='#c6ca49')
        plt.scatter(x_points12, y_points12, marker='x', color='r')

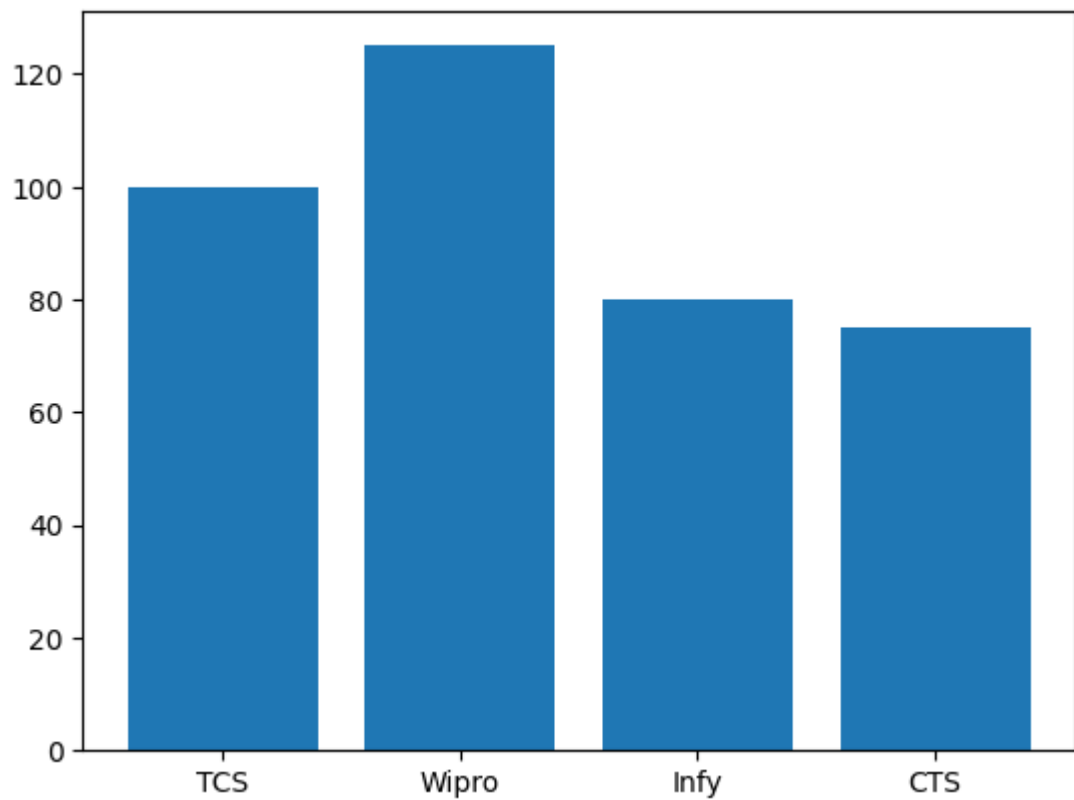
        plt.xlabel("Time")
        plt.ylabel("Sales")
        plt.title("A line chart of sales")
        plt.show()
```



Bars

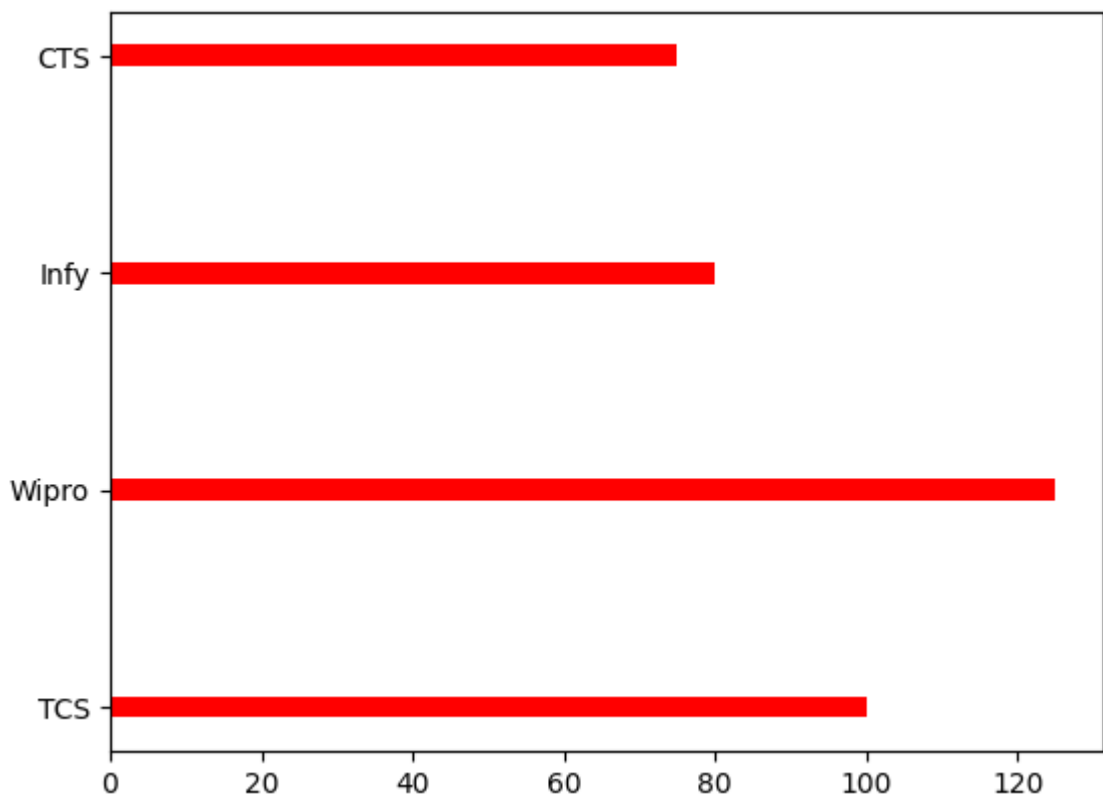
```
In [ ]: x = np.array(["TCS", "Wipro", "Infy", "CTS"])
        y = np.array([100, 125, 80, 75])

        plt.bar(x, y)
        plt.show()
```



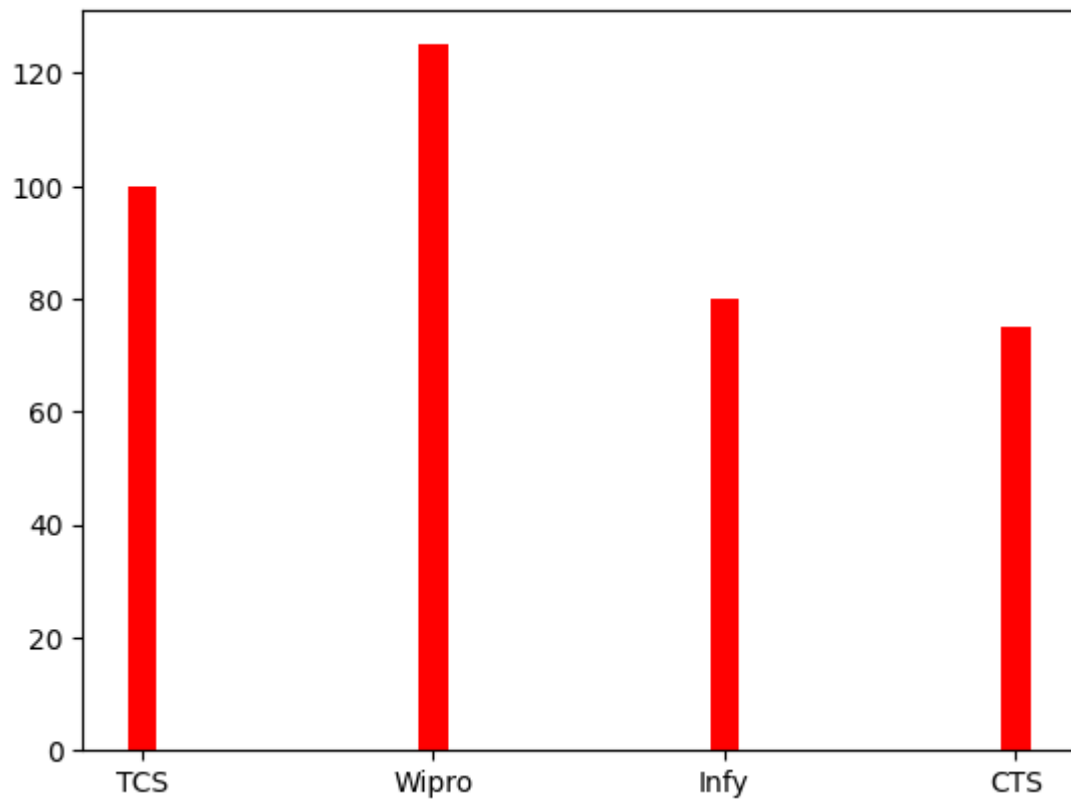
```
In [ ]: x = np.array(["TCS", "Wipro", "Infy", "CTS"])
        y = np.array([100, 125, 80, 75])

        plt.barh(x, y, color='red', height=0.1)
        plt.show()
```



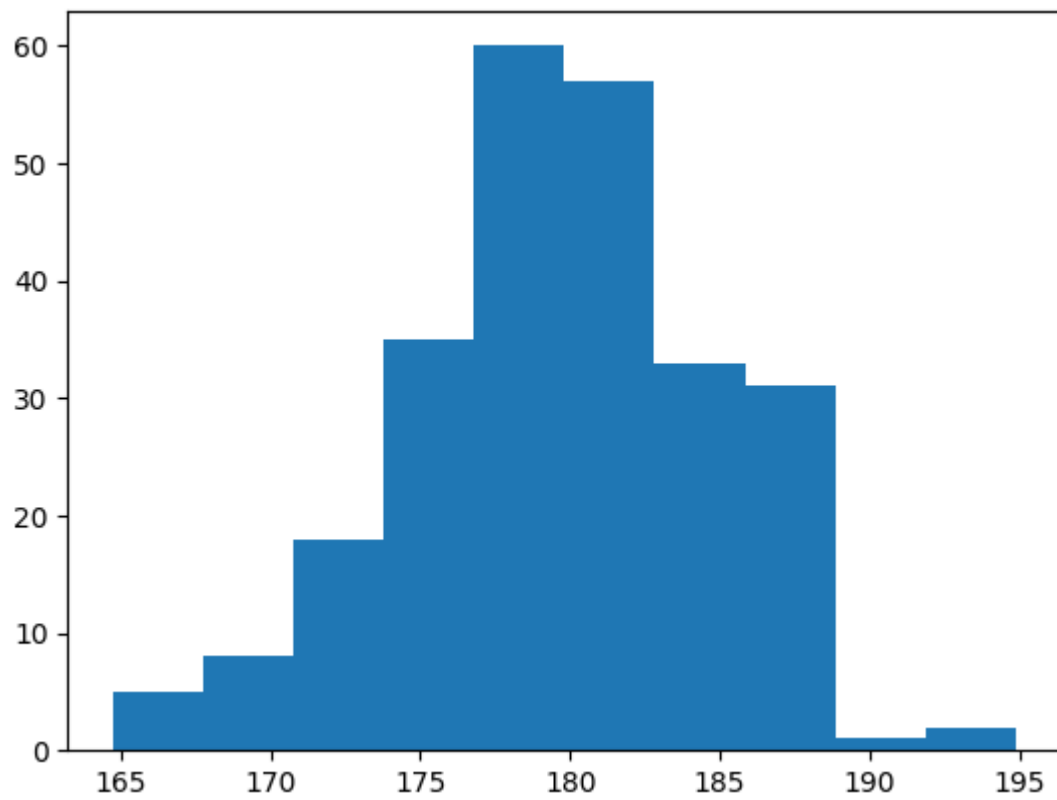
```
In [ ]: x = np.array(["TCS", "Wipro", "Infy", "CTS"])
y = np.array([100, 125, 80, 75])

plt.bar(x, y, color='red', width=0.1)
plt.show()
```



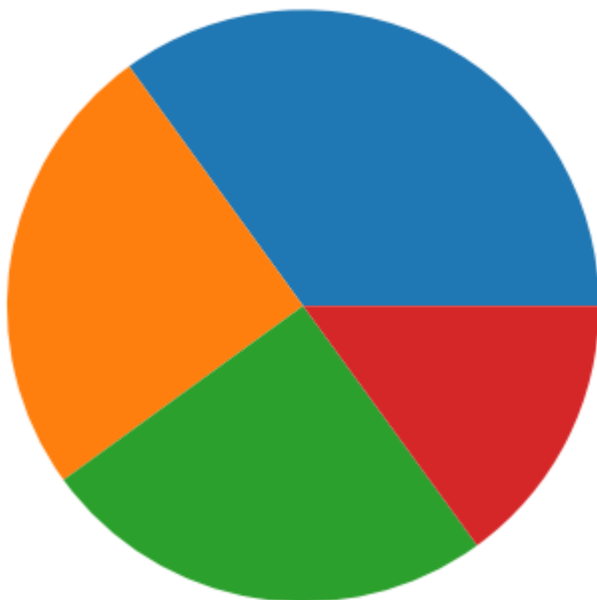
Histogram

```
In [ ]: x = np.random.normal(180, 5, 250)
plt.hist(x)
plt.show()
```

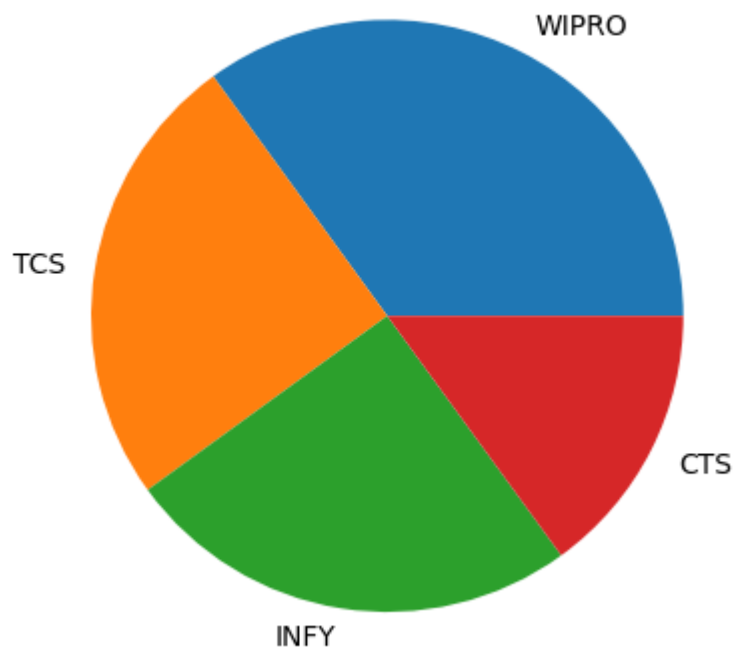


Piecharts

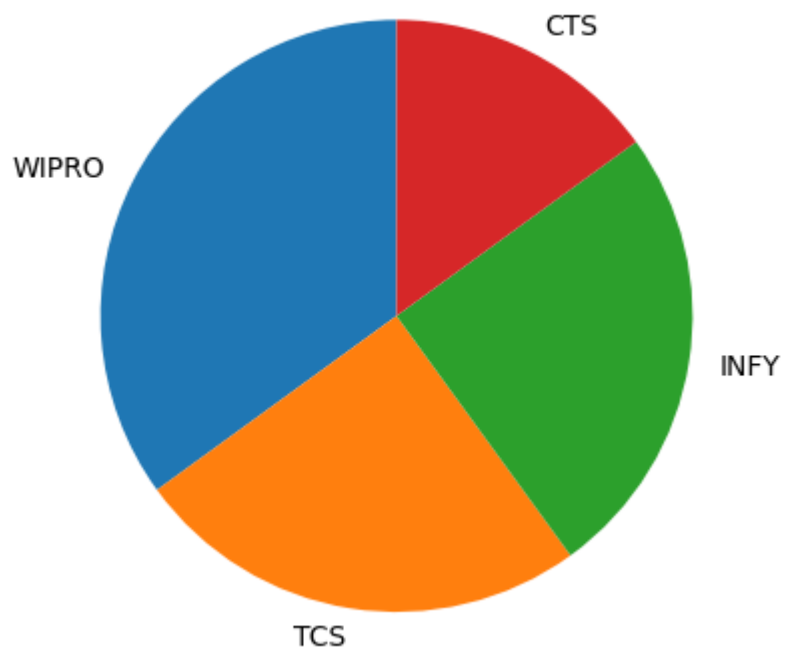
```
In [ ]: y = np.array([35,25,25,15])  
plt.pie(y)  
plt.show()
```



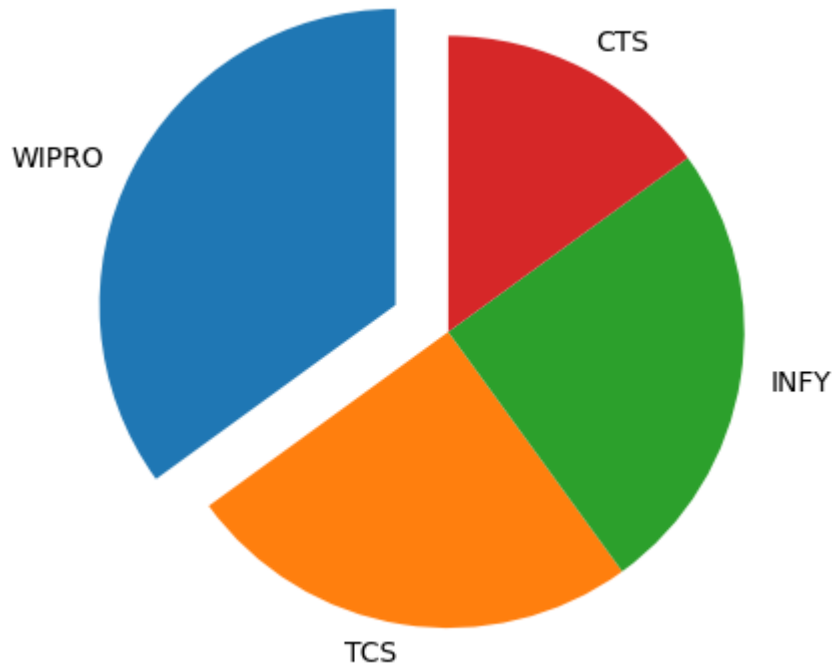
```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
plt.pie(y, labels=mylabels)
plt.show()
```



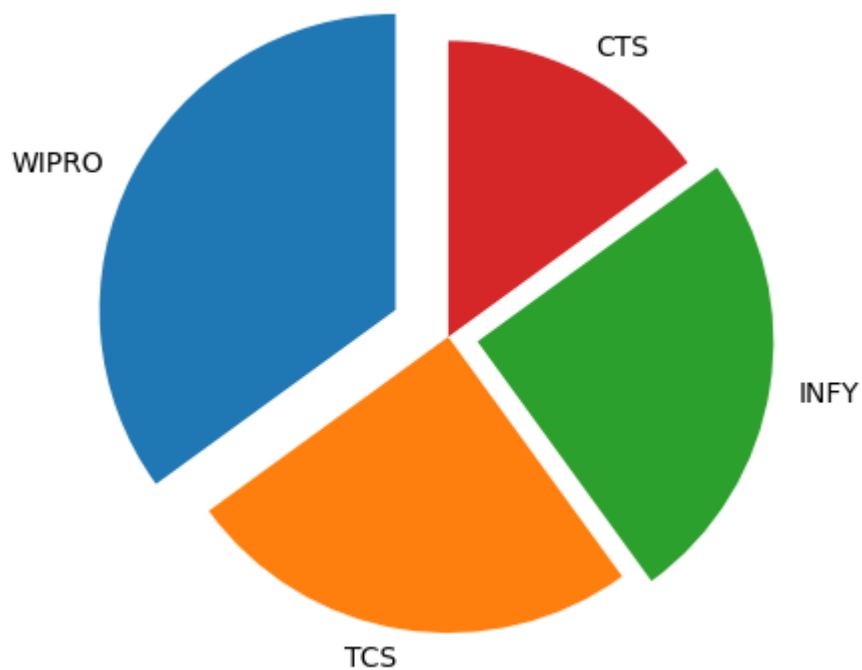
```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
plt.pie(y, labels=mylabels, startangle=90)
plt.show()
```



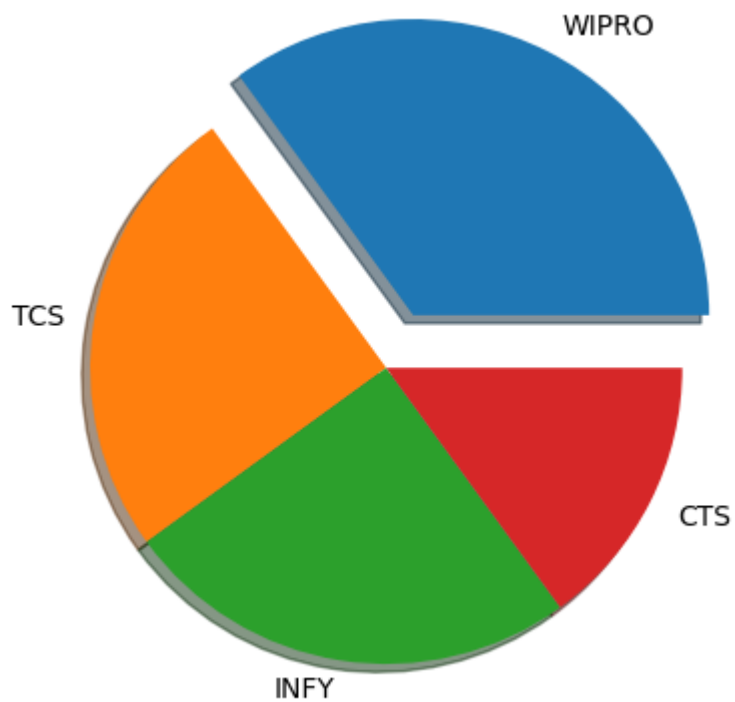
```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
myexplode=[0.2,0,0,0]
plt.pie(y, labels=mylabels, startangle=90, explode=myexplode)
plt.show()
```



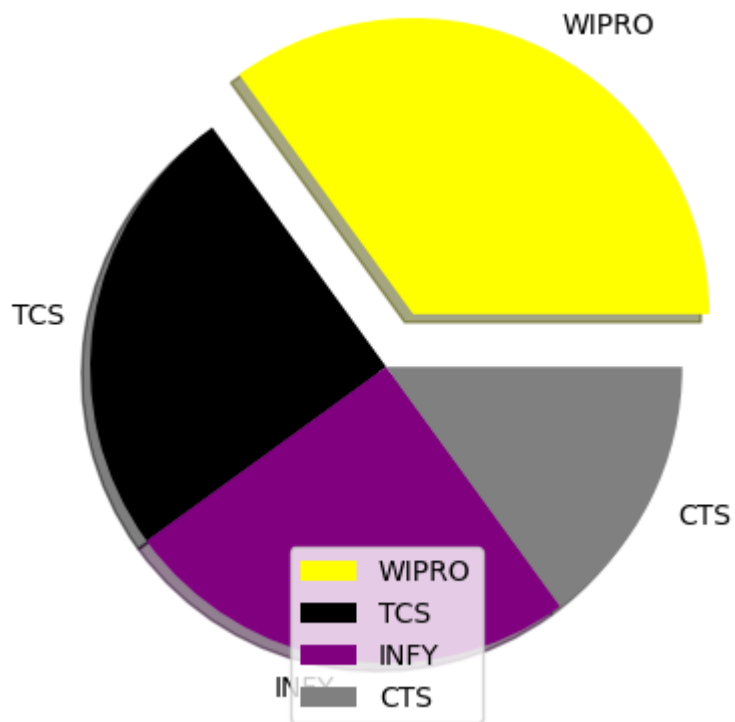
```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
myexplode=[0.2,0,0.1,0]
plt.pie(y, labels=mylabels, startangle=90, explode=myexplode)
plt.show()
```



```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
myexplode=[0.2,0,0,0]
plt.pie(y, labels=mylabels,explode=myexplode, shadow=True)
plt.show()
```



```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
myexplode=[0.2,0,0,0]
mycolors = ["yellow","black","purple","grey"]
plt.pie(y, labels=mylabels,explode=myexplode, shadow=True, colors=mycolor
plt.legend()
plt.show()
```



```
In [ ]: y = np.array([35,25,25,15])
mylabels = ["WIPRO","TCS","INFY","CTS"]
myexplode=[0.2,0,0,0]
mycolors = ["yellow","black","purple","grey"]
plt.pie(y, labels=mylabels,explode=myexplode, shadow=True, colors=mycolor)
plt.legend(title="company shares")
plt.show()
```

